NTNU
Norwegian University of
Science and Technology

Department of Electronic Systems

Examination paper for TFE 4164 Nanoelectronics 1

Academic contact during examination: Jostein Grepstad

Phone: 47418251

Examination date: December 14th, 2018

Examination time (from-to): 09.00 - 13.00

Permitted examination support material: C

Specified printed matter:

- Karl Rottmann, Formelsammlung/Collection of mathematical formulas.

- Specified simple calculator

Other information:

This exam has three problem sets: Problem set 1 is "multiple choice", tear off and use the attached answer sheet (final page).

The exam may be answered in English or Norwegian (*nynorsk* or *bokmål*), at your own preference.

The three problem sets are weighted as shown. As for the multiple-choice section, a correct answer is awarded +2 pts and each incorrect answer is awarded -1 pt. Zero, one, two, or three alternatives may be checked. There is only one correct answer to each question.

Language: English

Number of pages (front page excluded): 10

Number of pages enclosed: 6

Informasjon om trykking av eksamensoppgave

Originalen er:

1-sidig X **2-sidig ☐**

sort/hvit ☐ **farger X**

skal ha flervalgskjema X

Checked by:

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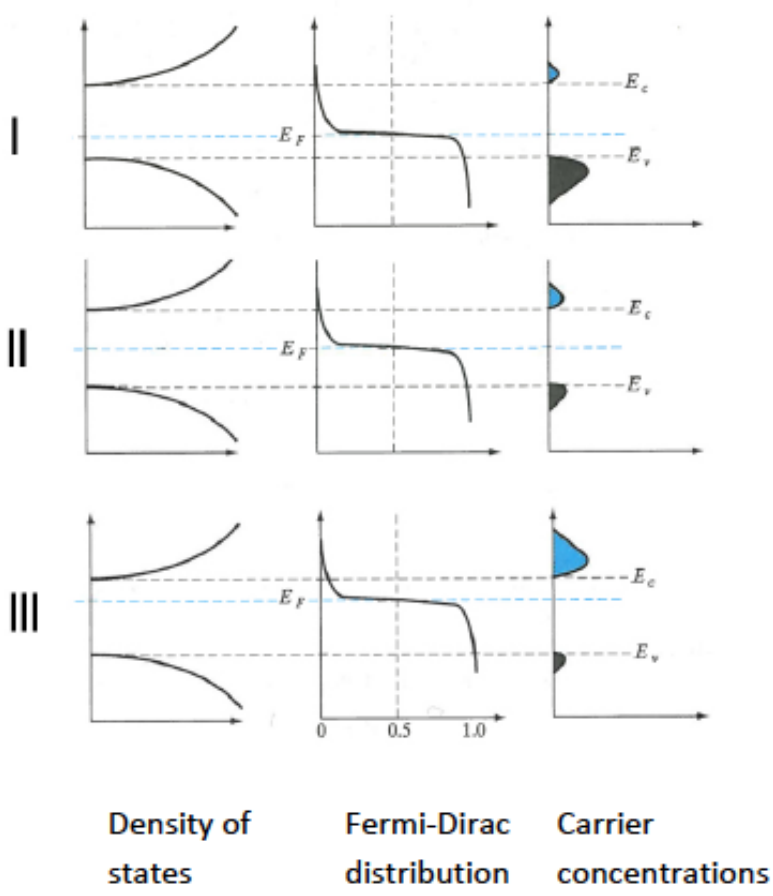
Date Signature

Problem set I [30 %]

MULTIPLE CHOICE

Q I-1

Figure 1



The doping conditions for the three cases depicted in Figure 1 above are,

(A) I: intrinsic II: n-type III: p-type

(B) I: p-type II: intrinsic III: n-type

(C) I: n-type II: intrinsic III: p-type

Q I-2

Which property is typically measured in the Haynes-Shockley experiment?

(A) the minority carrier mobility

(B) the majority carrier mobility

(C) the minority carrier density

Q I-3

The electron mobility μ_n in a semiconductor is limited by several scattering mechanisms, which result in different temperature dependencies for the mobility. If $\mu_n \propto T^{3/2}$ (T – the absolute temperature in K), the mobility is dominated by,

(A) lattice (phonon) scattering

(B) electron-electron scattering

(C) impurity scattering

Q I-4

In a p-n junction in thermal equilibrium there is,

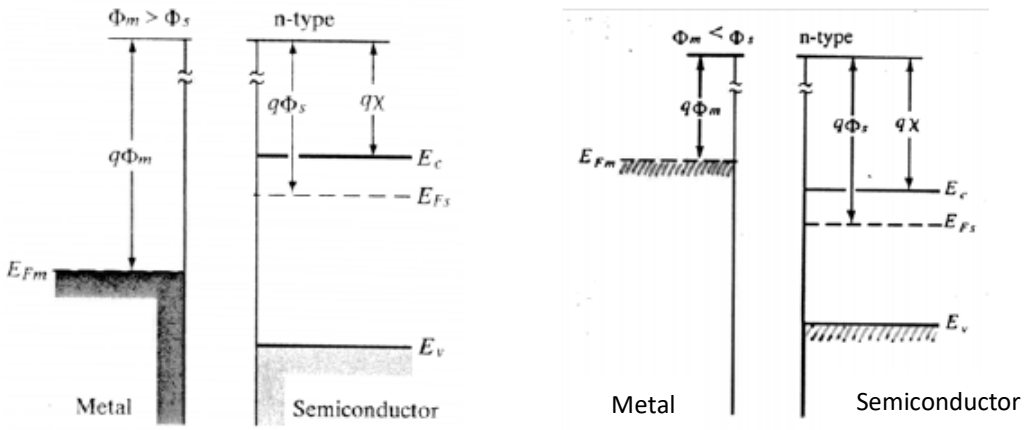
(A) zero diffusion current and a finite drift current

(B) zero drift current and a finite diffusion current

(C) finite drift and diffusion currents of equal magnitude and opposite sign (direction)

Q I-5

Figure 2



1 **2**

Figure 2 above shows band diagrams of two different metals and semiconductors, depicted before joining the two to form a metal–semiconductor junctions. Assuming no interface traps (surfacs states at the interface), which statement is correct?

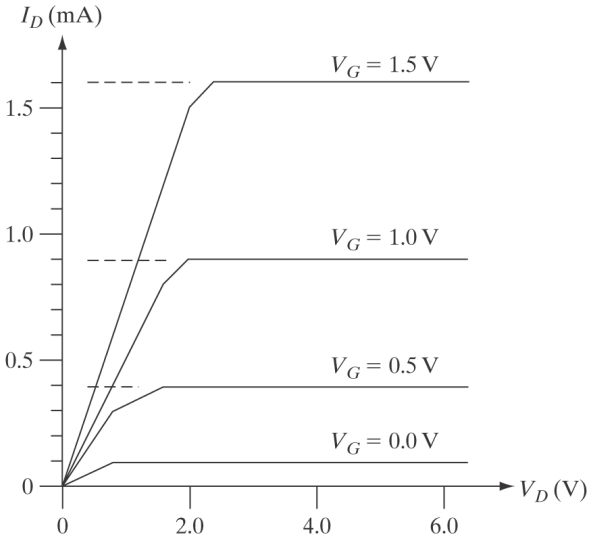
(A) Both **1** and **2** will result in Schottky contacts

(B) **1** will result in a Schottky contact, and **2** will result in an ohmic contact

(C) **1** will result in an ohmic contact, and **2** will result in a Schottky contact

Q I-6

Figure 3



The MOSFET I_D–V_D characteristics in Figure 3 above were measured for,

(A) an *enhancement mode* (i.e., normally off) *n-channel* MOSFET

(B) an *enhancement mode* (i.e., normally off) *p-channel* MOSFET

(C) a *depletion mode* (i.e., normally on) *n-channel* MOSFET

Q I-7

The subthreshold slope of a MOSFET [mV/decade] is decreased (i.e., improved) by,

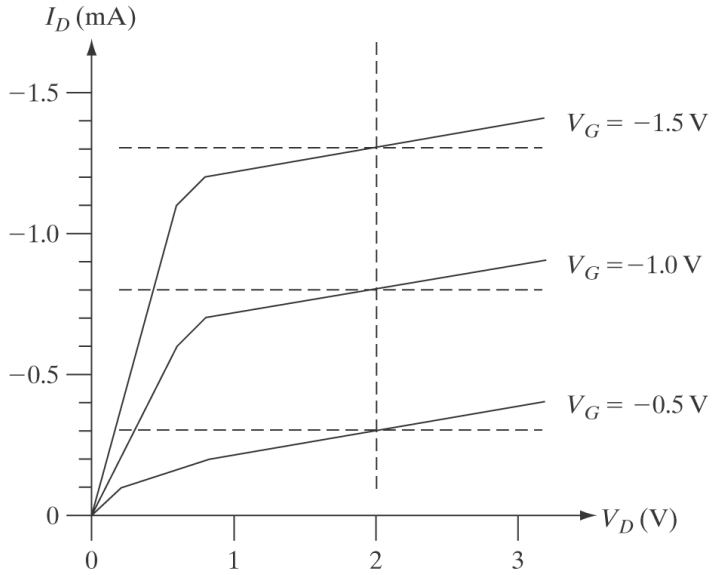
(A) decreasing the oxide thickness and increasing the substrate doping

(B) decreasing the oxide thickness and decreasing the substrate doping

(C) increasing the oxide thickness and decreasing the substrate doping

Q I-8

Figure 4



The MOSFET I_D–V_D characteristics in Figure 4 above were measured for an enhancement mode (i.e., normally off) device. They are the I_D–V_D characteristics of,

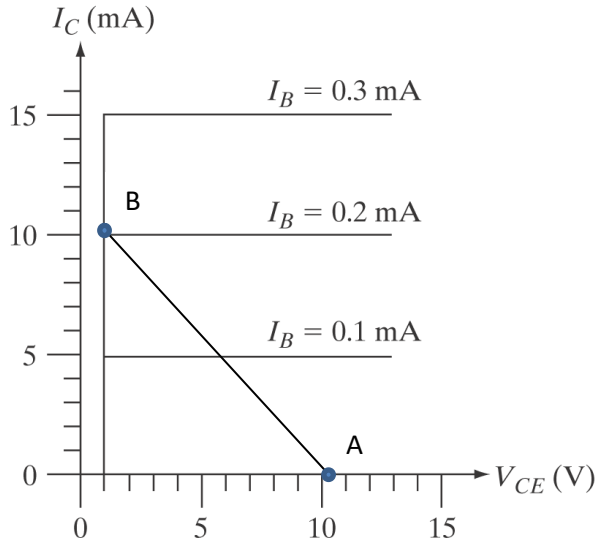
(A) a *long channel*, *p-channel* MOSFET

(B) a *short channel*, *p-channel* MOSFET

(C) a *short channel*, *n-channel* MOSFET

Q I-9

Figure 5



The diagram in Figure 5 above shows output I–V characteristics and load line for a bipolar junction transistor (BJT) wired up in common-emitter configuration. When tuning the base current i_B between 0 mA and 0.2 mA (i.e., between positions A and B on the load line),

(A) the BJT switches between *normal active* mode and *inverted active* mode

(B) the BJT switches between *cut-off* mode and *saturation* mode

(C) the BJT switches between *normal active* mode and *cut-off* mode

Q I-10

Decreasing the base width of a (BJT) will (assuming the device remains unchanged otherwise),

(A) increase the base transport factor, B and increase the emitter efficiency, γ

(B) increase the base transport factor, B and decrease the emitter efficiency, γ

(C) decrease the base transport factor, B and leave the emitter efficiency, γ unchanged

Q I-11

Decreasing the base width of a bipolar junction transistor (BJT) will (assuming the device remains unchanged otherwise),

(A) increase the magnitude of the Early voltage, V_A

(B) leave the magnitude of the Early voltage, V_A unchanged

(C) decrease the magnitude of the Early voltage, V_A

Q I-12

A silicon photovoltaic solar cell with *fill factor FF=0.8* delivers a maximum power of 28 W under full illumination. The open-circuit voltage V_{oc} and short-circuit current I_{sc} of this solar cell is,

(A) V_{oc} = 1.4 V and I_{sc} = 25 mA

(B) V_{oc} = 1.0 V and I_{sc} = 28 mA

(C) V_{oc} = 0.7 V and I_{sc} = 50 mA

Problem set II

p-n JUNCTION [35 pts]

Q II-1

Consider a one-sided abrupt p^+-n junction with a short (n-type) base region, i.e., a neutral base region, l shorter than the hole diffusion length, L_p

- Show that the current passing through this p^+-n junction can be expressed (to 1st order) on the form,

$$I = qAn_i^2 \frac{D_p}{N_D l}\left(e^{\frac{qV_a}{kT}} - 1 \right)$$

where A is the cross-section of the device, and V_a is the applied bias voltage.

Hint: Use the steady state diffusion equation to find an expression for the injected excess carrier concentration in the base region, $\delta p(x_n)$, where the conventional boundary condition of $\delta p(\infty) = 0$ for a p^+-n junction with base length $l >> L_p$ has been replaced by $\delta p(l) = 0$.

Use series expansion: $\tanh(x) = x - \frac{x^2}{3} + \frac{2x^5}{15} - \cdots$

Q II-2

Assume a Si p^+-n junction with cross section $A=10^{-4}$ cm^2 and a neutral base region of length $l < L_p$. The n-side dopant concentration is $N_D = 10^{17}$ cm^{-3}, while the p^+-side is heavily doped ($N_A >> N_D$) so that the Fermi level on this side coincides with the valence band edge, $E_F = E_V$.

- Sketch the energy band diagram $E_C(x)$ and $E_V(x)$ and the space charge carrier density $\rho(x)$ for this p^+-n junction in thermal equilibrium.

Assume $l = 2$ μm at zero bias voltage ($V_a = 0$), a hole diffusivity $D_p = 20$ cm^{2}s^{-1}, and ambient temperature, T = 300 K. The intrinsic charge carrier density for Si is $n_i = 1.5\times10^{10}$ cm^{-3}, and the bandgap is $E_g = 1.12$ eV.

- Calculate the forward bias voltage V_a for a diode current I = 1mA through the p^+-n junction. We assume an intrinsic Fermi level $E_i = ½(E_V + E_C)$.

Hint: One numerical iteration required

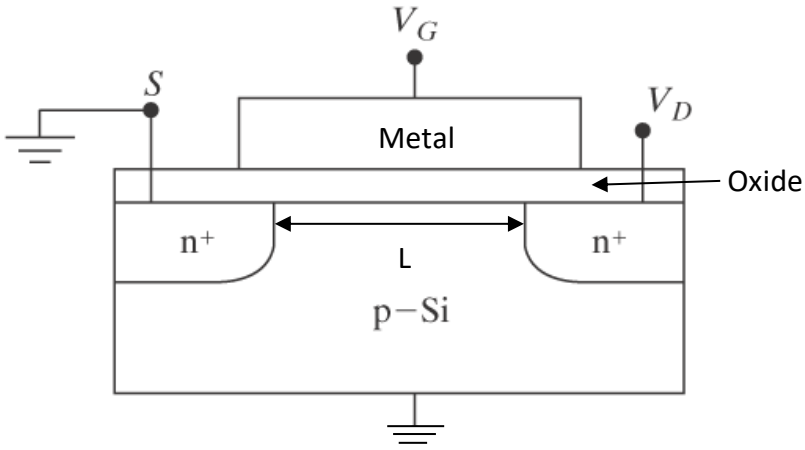
Problem set III [35 pts]

MOSFET

Q III-1

Figure 1 below shows the cross-section of an n-channel MOSFET in Si – with numerical values for the substrate doping, N_A, the gate oxide thickness, d and relative dielectric constant, ε_r.

Figure 1



$N_A = 4\times10^{15}$ cm^{-3}; d = 100 nm; $\varepsilon_r = 3.8$; $n_i = 1.5\times10^{10}$ cm^{-3}; $E_g = 1.12$ eV.

Assume an idéal MOS structure with flat bands in the semiconductor for zero applied gate voltage, $V_G = 0$.

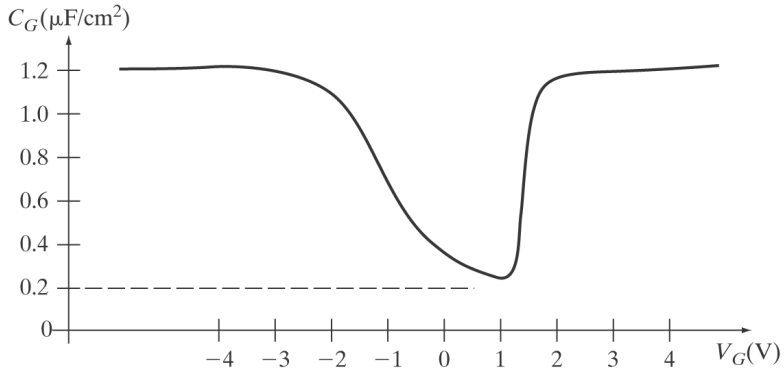
- Find the energy difference $q\varphi_F = E_i - E_{Fs}$ [eV] in the Si substrate in thermal equilibrium.
- At what potential φ_s on the semiconductor–oxide interface does this transistor enter *strong inversion*?
- Derive the depletion charge Q_d [C/cm^2] at the entrance of strong inversion for this device.
- Find the threshold voltage V_T for this transistor.
- Comment on material properties in a real MOS capacitor structure which will cause departure from flat bands in the semiconductor for zero gate voltage ($V_G = 0$).

Note: Except for the last bullet point, a numerical answer is required!

Q III-2

Figure 2 below shows the low-frequency gate capacitance vs. gate voltage relation for an idéal metal gate, n-channel MOSFET.

Figure 2



- Retrace this C_G–V_G characteristic, and label the approximate region or voltage corresponding to,

  weak inversion (1)

  flat bands (2)

  strong inversion (3)

  accumulation (4)

  threshold (5)

  depletion (6)

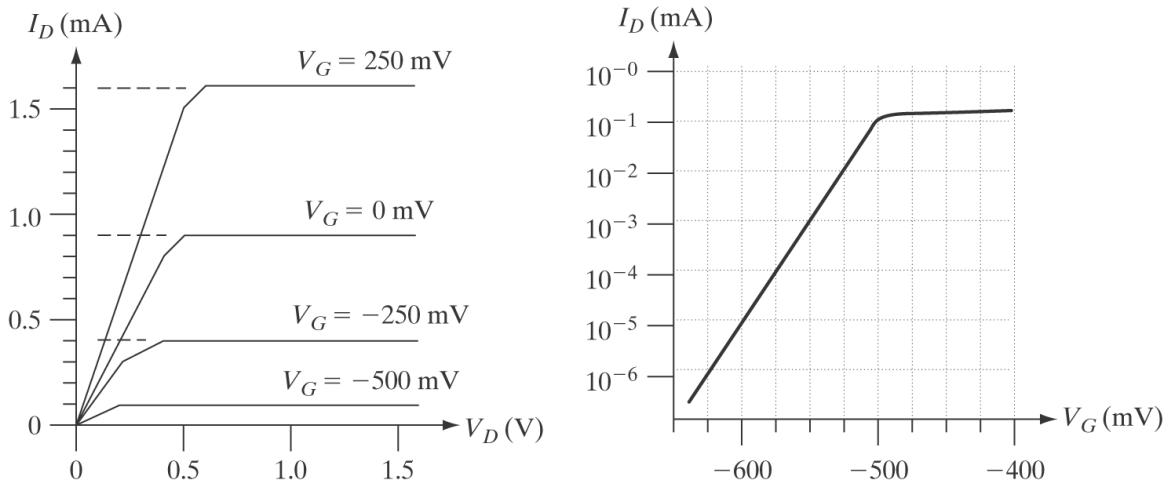
- What is the oxide capacitance, C_i, and the minimum depletion capacitance, $C_{d\,min}$, per unit area [$\mu F/cm^2$] for this MOSFET?

- Show how this C_G–V_G characteristic is modified under high frequency operation.

Q III-3

Figure 3 below show falsified (*fake*) I_D–V_D and I_D–V_G characteristics of an n-channel MOSFET, allegedly fabricated and measured at 300 K in a device fabrication course.

Figure 3



- If these were actual, measured I–V characteristics, would the MOSFET be a *depletion mode* (normally on) or an *enhancement mode* (normally off) device? Explain why!
- If these were actual, measured I–V characteristics, would this MOSFET be a "*long channel*" or a "*short channel*" device? Explain why!
- If these were actual, measured I–V characteristics, what would be the "*subthreshold slope/swing*" S [mV/decade] of this MOSFET?
- The falsification was eventually disclosed in this case. What physical aspect of these MOSFET I–V characteristics (besides the less than smooth curves) should draw suspicion?

  Hint: Concentrate on the I_D–V_G characteristics in Figure 3 above.

Physical constants and conversion factors

Avogadro's number	$N_A = 6.02 \times 10^{23}$ molecules/mole
Boltzmann's constant	$k = 1.38 \times 10^{-23}$ J/K
	$= 8.62 \times 10^{-5}$ eV/K
Electronic charge (magnitude)	$q = 1.60 \times 10^{-19}$ C
Electronic rest mass	$m_0 = 9.11 \times 10^{-31}$ kg
Permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-14}$ F/cm
	$= 8.85 \times 10^{-12}$ F/m
Planck's constant	$h = 6.63 \times 10^{-34}$ J-s
	$= 4.14 \times 10^{-15}$ eV-s
Room temperature value of kT	$kT = 0.0259$ eV
Speed of light	$c = 2.998 \times 10^{10}$ cm/s

	Prefixes:		
1 Å (angstrom) = 10^{-8} cm	milli-,	m-	$= 10^{-3}$
1 μm (micron) = 10^{-4} cm	micro-,	μ-	$= 10^{-6}$
1 nm = 10 Å = 10^{-7}cm	nano-,	n-	$= 10^{-9}$
2.54 cm = 1 in.	pico-,	p-	$= 10^{-12}$
1 eV = 1.6×10^{-19} J	kilo-,	k-	$= 10^{3}$
	mega-,	M-	$= 10^{6}$
	giga-,	G-	$= 10^{9}$

A wavelength λ of 1 μm corresponds to a photon energy of 1.24 eV.

Formulas TFE-4164

SEMICONDUCTOR PHYSICS

Electron Momentum: $\mathrm{p} = m^*\mathrm{v} = \hbar\mathbf{k} = \dfrac{h}{\lambda}$ Planck: $E = h\nu = \hbar\omega$

Kinetic: $E = \dfrac{1}{2}m^*\mathrm{v}^2 = \dfrac{1}{2}\dfrac{\mathrm{p}^2}{m^*} = \dfrac{\hbar^2}{2m^*}\mathbf{k}^2$ (3-4) Effective mass: $m^* = \dfrac{\hbar^2}{d^2E/d\mathbf{k}^2}$ (3-3)

Total electron energy = $P.E. + K.E. = E_c + E(\mathbf{k})$

Fermi-Dirac e^- distribution: $f(E) = \dfrac{1}{e^{(E-E_f)/kT}+1} \cong e^{(E_f-E)/kT}$ for $E \gg E_F$ (3-10)

Equilibrium: $n_0 = \displaystyle\int_{E_c}^{\infty} f(E)N(E)dE = N_c f(E_c) = N_c e^{-(E_c-E_f)/kT}$ (3-15)

$N_c = 2\left(\dfrac{2\pi m_n^* kT}{h^2}\right)^{3/2}$ $N_v = 2\left(\dfrac{2\pi m_p^* kT}{h^2}\right)^{3/2}$ (3-16), (3-20)

$p_0 = N_v[1 - f(E_v)] = N_v e^{-(E_f-E_v)/kT}$ (3-19)

$n_i = N_c e^{-(E_c-E_i)/kT}$, $p_i = N_v e^{-(E_i-E_v)/kT}$ (3-21)

$n_i = \sqrt{N_c N_v}\, e^{-E_g/2kT} = 2\left(\dfrac{2\pi kT}{h^2}\right)^{3/2}(m_n^* m_p^*)^{3/4}e^{-E_g/2kT}$ (3-23), (3-26)

Equilibrium: $n_0 = n_i e^{(E_f-E_i)/kT}$, $p_0 = n_i e^{(E_i-E_f)/kT}$ (3-25) $n_0 p_0 = n_i^2$ (3-24)

Steady state: $n = N_c e^{-(E_c-F_n)/kT} = n_i e^{(F_n-E_i)/kT}$, $p = N_v e^{-(F_p-E_v)/kT} = n_i e^{(E_i-F_p)/kT}$ (4-15) $np = n_i^2 e^{(F_n-F_p)/kT}$ (5-38)

$\mathscr{E}(x) = -\dfrac{d\mathscr{V}(x)}{dx} = \dfrac{1}{q}\dfrac{dE_i}{dx}$ (4-26)

Poisson: $\dfrac{d\mathscr{E}(x)}{dx} = -\dfrac{d^2\mathscr{V}(x)}{dx^2} = \dfrac{\rho(x)}{\epsilon} = \dfrac{q}{\epsilon}(p - n + N_d^+ - N_a^-)$ (5-14)

$\mu \equiv \dfrac{q\bar{t}}{m^*}$ (3-40a) Drift: $\mathrm{v}_d \cong \dfrac{\mu\mathscr{E}}{1 + \mu\mathscr{E}/\mathrm{v}_s}\begin{cases} = \mu\mathscr{E} \text{ (low fields, ohmic)} \\ = \mathrm{v}_s \text{ (high fields, saturated vel.)} \end{cases}$ (Fig. 6-9)

Drift current density: $\dfrac{I_x}{A} = J_x = q(n\mu_n + p\mu_p)\mathscr{E}_x = \sigma\mathscr{E}_x$ (3-43)

$$J_n(x) = q\mu_n n(x)\mathscr{E}(x) + qD_n \frac{dn(x)}{dx}$$

Conduction Current: drift diffusion (4–23)

$$J_p(x) = q\mu_p p(x)\mathscr{E}(x) - qD_p \frac{dp(x)}{dx}$$

$$J_{\text{total}} = J_{\text{conduction}} + J_{\text{displacement}} = J_n + J_p + C\frac{dV}{dt}$$

Continuity: $\frac{\partial p(x,t)}{\partial t} = \frac{\partial \delta p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} - \frac{\delta p}{\tau_p}$ $\frac{\partial \delta n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} - \frac{\delta n}{\tau_n}$ (4–31)

For steady state diffusion: $\frac{d^2\delta n}{dx^2} = \frac{\delta n}{D_n\tau_n} \equiv \frac{\delta n}{L_n^2}$ $\frac{d^2\delta p}{dx^2} = \frac{\delta p}{L_p^2}$ (4–34)

Diffusion length: $L \equiv \sqrt{D\tau}$ Einstein relation: $\frac{D}{\mu} = \frac{kT}{q}$ (4–29)

p-n JUNCTIONS

Equilibrium: $V_0 = \frac{kT}{q}\ln\frac{p_p}{p_n} = \frac{\text{kT}}{\text{q}}\ln\frac{N_a}{n_i^2/N_d} = \frac{\text{kT}}{\text{q}}\ln\frac{N_aN_d}{n_i^2}$ (5-8)

$\frac{p_p}{p_n} = \frac{n_n}{n_p} = e^{qV_0/kT}$ (5-10) $W = \left[\frac{2\epsilon(V_0 - V)}{q}\left(\frac{N_a + N_d}{N_aN_d}\right)\right]^{1/2}$ (5-57)

One-sided abrupt p^+-n: $x_{n0} = \frac{WN_a}{N_a + N_d} \simeq W$ (5-23) $V_0 = \frac{qN_dW^2}{2\epsilon}$

$\Delta p_n = p(x_{n0}) - p_n = p_n(e^{qV/kT} - 1)$ (5-29)

$\delta p(x_n) = \Delta p_n e^{-x_n/L_p} = p_n(e^{qV/kT} - 1)e^{-x_n/L_p}$ (5-31b)

Ideal diode: $I = qA\left(\frac{D_p}{L_p}p_n + \frac{D_n}{L_n}n_p\right)(e^{qV/kT} - 1) = I_0(e^{qV/kT} - 1)$ (5-36)

Non-ideal: $I = I_0'(e^{qV/\mathfrak{n}kT} - 1)$ ($\mathfrak{n}$ = 1 to 2) (5-74)

With light: $I_{\text{op}} = qAg_{\text{op}}(L_p + L_n + W)$ (8-1)

Capacitance: $C = \left|\frac{dQ}{dV}\right|$ (5-55)

Junction Depletion: $C_j = \epsilon A\left[\frac{q}{2\epsilon(V_0 - V)}\frac{N_d N_a}{N_d + N_a}\right]^{1/2} = \frac{\epsilon A}{W}$ (5-62)

Stored charge exp. hole dist.: $Q_p = qA\int_0^\infty \delta p(x_n)dx_n = qA\Delta p_n\int_0^\infty e^{-x_n/L_p}dx_n = qAL_p\Delta p_n$ (5-39)

$I_p(x_n = 0) = \frac{Q_p}{\tau_p} = qA\frac{L_p}{\tau_p}\Delta p_n = qA\frac{D_p}{L_p}p_n(e^{qV/kT} - 1)$ (5-40)

$G_s = \frac{dI}{dV} = \frac{qAL_p p_n}{\tau_p}\frac{d}{dV}(e^{qV/kT}) = \frac{q}{kT}I$ (5-67c)

Long p^+-n: $i(t) = \frac{Q_p(t)}{\tau_p} + \frac{dQ_p(t)}{dt}$ (5-47)

MOS-n CHANNEL

Oxide: $C_i = \frac{\epsilon_i}{d}$ Depletion: $C_d = \frac{\epsilon_s}{W}$ MOS: $C = \frac{C_i C_d}{C_i}$ (6-36)

Threshold: $V_T = \underbrace{\Phi_{ms} - \frac{Q_i}{C_i}}_{\text{Flat band}} - \frac{Q_d}{C_i} + 2\phi_F$ (6-38)

Inversion: $\phi_s\,(\text{inv.}) = 2\phi_F = 2\frac{kT}{q}\ln\frac{N_a}{n_i}$ (6-15) $W = \left[\frac{2\epsilon_s\phi_s}{qN_a}\right]^{1/2}$ (6-30)

$Q_d = -qN_aW_m = -2(\epsilon_s qN_a\phi_F)^{1/2}$ (6-32) At V_{FB}: $C_{FB} = \frac{C_i C_{\text{debye}}}{C_i + C_{\text{debye}}}$

Debye screening length: $L_D = \sqrt{\frac{\epsilon_s kT}{q^2 p_0}}$ (6-25) $C_{\text{debye}} = \frac{\epsilon_s}{L_D}$ (6-40)

Substrate bias: $\Delta V_T \simeq \frac{\sqrt{2\epsilon_s qN_a}}{C_i}(-V_B)^{1/2}$ (n channel) (6-63)

$$I_D = \frac{\bar{\mu}_n Z C_i}{L}\left\{(V_G - V_{FB} - 2\phi_F - \tfrac{1}{2}V_D)V_D - \frac{2}{3}\frac{\sqrt{2\epsilon_s q N_a}}{C_i}[(V_D + 2\phi_F)^{3/2} - (2\phi_F)^{3/2}]\right\} \quad (6\text{-}50)$$

$$I_D \simeq \frac{\bar{\mu}_n Z C_i}{L}[(V_G - V_T)V_D - \tfrac{1}{2}V_D^2] \quad (6\text{-}49)$$

Saturation: $I_D(\text{sat.}) \simeq \tfrac{1}{2}\bar{\mu}_n C_i \frac{Z}{L}(V_G - V_T)^2 = \frac{Z}{2L}\bar{\mu}_n C_i V_D^2(\text{sat.}) \quad (6\text{-}53)$

$$g_m = \frac{\partial I_D}{\partial V_G} \quad ; \quad g_m(\text{sat.}) = \frac{\partial I_D(\text{sat.})}{\partial V_G} \simeq \frac{Z}{L}\bar{\mu}_n C_i(V_G - V_T) \quad (6\text{-}54)$$

For short L: $I_D \simeq Z C_i(V_G - V_T)v_s \quad (6\text{-}60)$

Subthreshold slope: $S = \frac{dV_G}{d(\log I_D)} = \frac{kT}{q}\ln 10\left[1 + \frac{C_d + C_{it}}{C_i}\right] \quad (6\text{-}66)$

BJT-p-n-p

$$I_{Ep} = qA\frac{D_p}{L_p}\left(\Delta p_E \operatorname{ctnh}\frac{W_b}{L_p} - \Delta p_C \operatorname{csch}\frac{W_b}{L_p}\right) \quad (7\text{-}18) \qquad \begin{aligned}\Delta p_E &= p_n(e^{qV_{EB}/kT} - 1)\\ \Delta p_C &= p_n(e^{qV_{CB}/kT} - 1)\end{aligned} \quad (7\text{-}8)$$

$$I_C = qA\frac{D_p}{L_p}\left(\Delta p_E \operatorname{csch}\frac{W_b}{L_p} - \Delta p_C \operatorname{ctnh}\frac{W_b}{L_p}\right)$$

$$I_B = qA\frac{D_p}{L_p}\left[(\Delta p_E + \Delta p_C)\tanh\frac{W_b}{2L_p}\right] \quad (7\text{-}19)$$

$$B = \frac{I_C}{I_{Ep}} = \frac{\operatorname{csch} W_b/L_p}{\operatorname{ctnh} W_b/L_p} = \operatorname{sech}\frac{W_b}{L_p} \simeq 1 - \left(\frac{W_b^2}{2L_p^2}\right) \quad (7\text{-}26)$$

(Base transport factor)

$$\gamma = \frac{I_{Ep}}{I_{En} + I_{Ep}} = \left[1 + \frac{L_p^n n_n \mu_n^p}{L_n^p p_p \mu_p^n}\tanh\frac{W_b}{L_p^n}\right]^{-1} \simeq \left[1 + \frac{W_b n_n \mu_n^p}{L_n^p p_p \mu_p^n}\right]^{-1} \quad (7\text{-}25)$$

(Emitter injection efficiency)

$$\frac{i_C}{i_E} = B\gamma \equiv \alpha \quad (7\text{-}3) \qquad \frac{i_C}{i_B} = \frac{B\gamma}{1 - B\gamma} = \frac{\alpha}{1 - \alpha} \equiv \beta \quad (7\text{-}6) \qquad \frac{i_C}{i_B} = \beta = \frac{\tau_p}{\tau_t} \quad (7\text{-}7)$$

(Common base gain) (Common emitter gain) (For $\gamma = 1$)

Answer sheet for "Multiple Choice" questions

(Tear off this form, check your answers, and hand in page together with the rest of your exam papers.)

Candidate number: _____

Alternative / Problem no.	A	B	C
I-1			
I-2			
I-3			
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